



Capstone Project 2

CMU-SE 451

Database Design Document

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Sign Language Detection System

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PROJECT INFORMATION

Project acronym	SLDS		
Project Title	Sign Language Detection System		
Start Date	15 - Feb - 2024	End Date	20 - May - 2024
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Document Approval

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1. Introduction

1.1. Purpose

- The main aim of the project is to create an effective communication tool between two groups of people with different communication approaches: sign language users and normal people.
- The project aims to reduce communication barriers and enhance the connection and integration of the deaf and mute community into society through technology.

1.2. Goal

The goal is to create database tables with the highest accuracy to support the website Sign Language Detection System.

1.3. Scope

- The project focuses on developing a video call communication system between sign language users and normal people.
- The scope of the project includes building the user interface, integrating real-time translation and gesture recognition technology, as well as ensuring system flexibility and security.

1.4. Data Storage Platforms

- Data of the application is stored in the PostgreSQL database
- In PostgreSQL, data structure is stored as tables the database objects that behave as containers for the data, in which the data will be logically organized in rows and columns format.

1.5. Definition, Acronyms, and Abbreviations

Table 1: *Definition, Acronyms, and Abbreviations*

Abbreviations	Description	Comment
PK/FK	Primary/ Foreign Key	Use to indicate a file is a Primary or Foreign key in a table
ERD	Entity Relationship Diagram	Show the relationship between entities in the system

1.6. Entity relationship chart

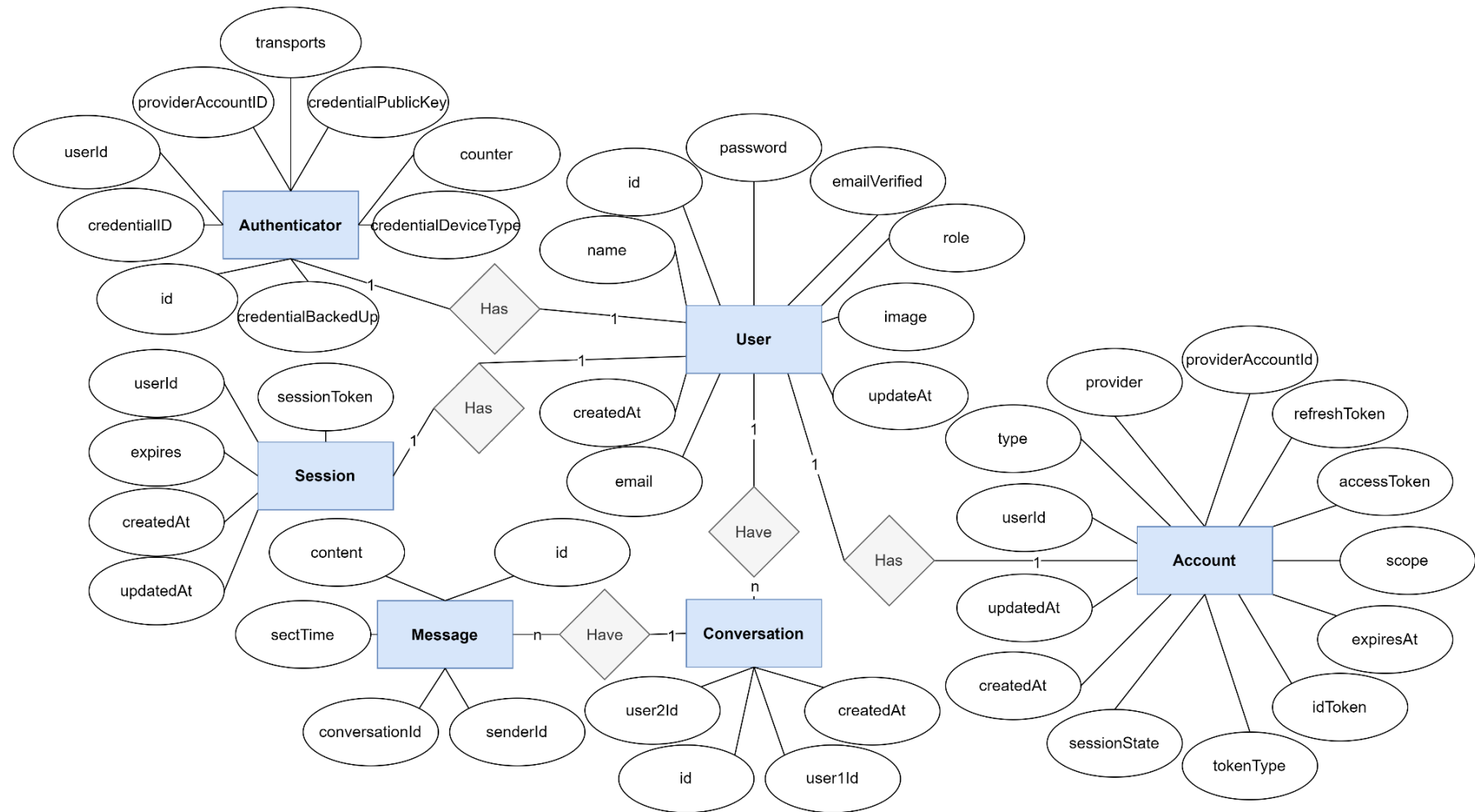


Figure 01: URs entity relationship chart

2. Relational Model

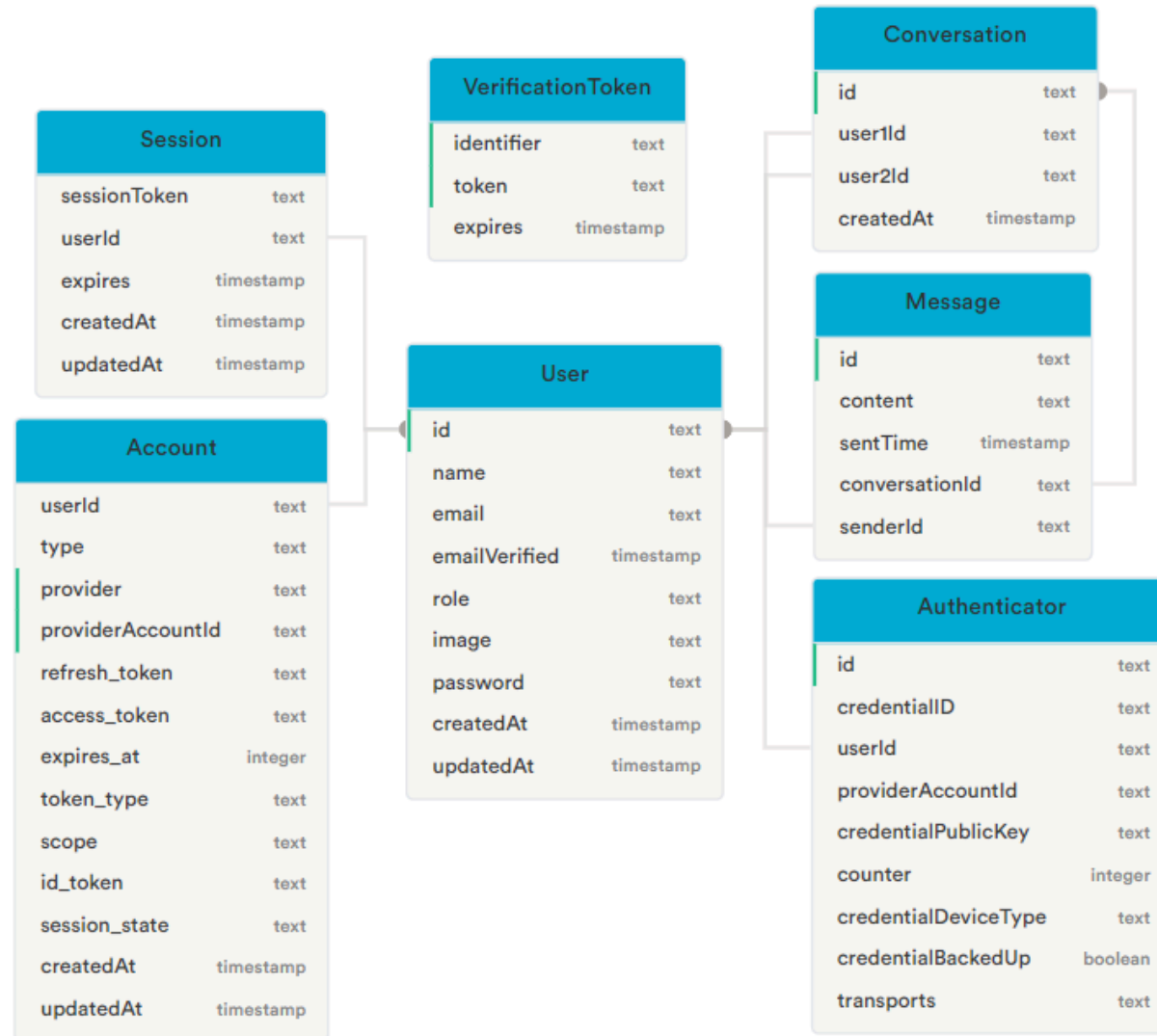


Figure 02: URs relational mode

3. Detail of the tables

3.1. Table Overview

Table 2: Table Overview

No	Table name	Short Description
1	User	This table contains the user's personal information such as name, email, and photo.
2	Account	This table stores user account information, including login information.
3	Session	This table manages user-session information.
4	VerificationToken	This table contains verification codes used to authenticate users.
5	Message	This table manages messages sent in the system.
6	Conversation	This table tracks conversations between users.
7	Authenticator	This table contains information about authenticating devices.

3.2. User table

ID	Field	Datatype	Constraint	Description
01	id	text	PK	A unique identifier for each user.
02	name	text		The user's name.
03	email	text		The user's email address.
04	emailVerified	timestamp		Timestamp indicating when the email was verified.
05	image	text		A path to the user's profile image.
06	role	text		The user's role.
07	password	text		The user's password.
08	createdAt	timestamp		Timestamp when the user was created.
09	updatedAt	timestamp		Timestamp when the user was last updated.

Table 3: User

3.3. Account table

Table 4: *Account*

ID	Field	Datatype	Constraint	Description
01	userId	text	FK	ID of the associated user.
02	type	text		Type of account.
03	provider	text		Account provider.
04	providerAccountId	text		ID assigned by the provider.
05	refreshToken	text		Refresh token for OAuth.
06	accessToken	text		Access token for OAuth.
07	expiresAt	integer		Expiration timestamp for tokens.
08	tokenType	text		Type of token.
09	scope	text		Scope of access.
10	idToken	text		ID token for OAuth.
11	sessionState	text		Session state for OAuth.
12	createdAt	timestamp		Timestamp when the account was created.
13	updatedAt	timestamp		Timestamp when the account was last updated.

3.4. Session table

Table 5: *Session*

ID	Field	Datatype	Constraint	Description
01	sessionToken	text		Token for user sessions.
02	userId	text	FK	ID of the associated user.
03	expires	timestamp		Expiration timestamp for the session.
04	createdAt	timestamp		Timestamp when the session was created.
05	updatedAt	timestamp		Timestamp when the session was last updated.

3.5. VerificationToken table

Table 6: *VerificationToken*

ID	Field	Datatype	Constraint	Description
01	identifier	text	PK	Identifier for verification.
02	token	text	PK	Verification token.
03	expires	timestamp		Expiration timestamp for the token.

3.6. Message table

Table 7: *Message*

ID	Field	Datatype	Constraint	Description
01	id	text	PK	Unique ID for messages.
02	content	text		The content of the message.
03	sentTime	timestamp		Timestamp when the message was sent.
04	conversationId	text	FK	ID of the associated conversation.
05	senderId	text	FK	ID of the message sender

3.7. Conversation table

Table 8: *Conversation*

ID	Field	Datatype	Constraint	Description
01	id	text	PK	Unique ID for conversations.
02	user1Id	text	FK	ID of the first user in the conversation.
03	user2Id	text	FK	ID of the second user in the conversation.
03	createdAt	timestamp		Timestamp when the conversation was created.

3.8. Authenticator table

Table 9: *Authenticator*

ID	Field	Datatype	Constraint	Description
01	id	text	PK	Unique ID for authenticators.
02	credentialId	text		ID for the associated credential.
03	userId	text	FK	ID of the associated user.
04	providerAccountId	text		ID from the provider.
05	credentialPublicKey	text		Public key for the credential.
06	counter	integer		Counter for authenticator usage.
07	credentialDeviceType	text		Device type for the credential.
08	CredentialBackedUp	boolean		Indicates if the credential is backed up.
09	transports	text		Transport methods